

MACKENZIE HALLER

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PROFILE

Healthcare and business analytics professional with experience supporting reporting, data validation, and operational analysis in healthcare and regulated environments. Comfortable working with large datasets, identifying trends, and communicating findings to cross-functional teams. Strong skills in SQL, Excel, and reporting tools, with a focus on clear communication.

EDUCATION

Master of Health Data Science (Honors) Feb 2024
University of New South Wales (UNSW)

B.S. Applied Mathematics, Minor Economics Sep 2015
California Polytechnic State University | San Luis Obispo, California

TECHNICAL SKILLS

Programming & Analytics: Python, R (Shiny, dplyr, quarto), SQL, SAS, Java, Excel (VBA)
Healthcare & Data Analytics: Population health reporting, clinical and operational metrics, clinical data analysis, data quality assurance, reproducible reporting workflows
Data Visualization & Applications: Power BI (DAX), Spotfire, Plotly, ggplot2, matplotlib, seaborn, Shiny, Streamlit
Machine Learning & Statistical Analysis: Regression, classification, survival analysis, tree-based models, LLM workflows (RAG & LoRa)
Data Infrastructure & Workflow: SQL Server, schema design, ETL pipelines, Azure, Docker, Git/GitHub, LLM data collection
GIS / Spatial Analytics: R (sf, leaflet, ggmap)
Communication & Stakeholder Engagement: Translating data insights into actionable summaries for clinical, operational, and leadership audiences
Workflow Enhancement: Process improvement & automation

EXPERIENCE

Data Science Officer Sep 2024 - Present
Royal Life Saving Non-Profit (Public Health)

- Conduct research and data analysis for national public health programs, supporting study protocols and evidence-based interventions.
- Perform data cleaning, validation, and consistency checks to ensure data quality across large, multi-source public health datasets.
- Designed database in SQL server and maintained Azure data pipelines, troubleshooting data flow issues and ensuring reliable execution.
- Developed R Shiny & Python tools to support data collection and real-time reporting and operational decision-making.
- Collaborated with internal partners on metrics and reporting requirements
- Created GIS-based tools and reproducible reports for national publications.
- Upskill team to use data tools like R and SQL to help enhance efficiency
- Ad hoc reporting and data analysis to help create campaigns and provide data to public for research
- Automate and enhance workflows to improve efficiency and support reproducibility
- Create dashboards for internal and external use
- Building data collection pipeline using LLM's

Data/Business Analyst Nov 2022 - Jan 2024
ASML | San Diego

- Designed and developed interactive dashboards integrating multiple complex data sources to perform cost analytics
- Analyzed demand, cost, and forecast trends to support operational and financial planning for future part production
- Collaborated with cross-functional teams to interpret data insights and translate them into actionable business and supply chain recommendations.
- Presented weekly and quarterly performance reports to senior leadership, including executive-level stakeholders, to support strategic decision-making.
- Delivered data-driven recommendations that helped reduce cost inefficiencies and improve part utilization and operational effectiveness.

Engineering/Data Analyst Mar 2020 - Nov 2022
San Diego Gas & Electric (SDG&E) | [Redacted Content]

Analyzed large-scale operational datasets to support regulatory reporting and internal data quality standards.
Authored daily operational reports, ensuring timely and accurate communication of findings.

Data Engineer / QA (Part-time) Jan 2022 - Jun 2022
GoodMonth Labs | Remote

Developed an NLP-based ML model for consumer messaging data.

Quality Assurance Analyst Aug 2019 - Mar 2020
Arrowhead General Insurance | [Redacted Content]

Designed and executed QA strategies for enterprise applications, improving release quality.

PROJECTS

UNSW Data Science Datathon Dec 2024
Collaborated with a 4-person team to analyze real-time EpiWatch epidemic data.

Resident Scheduling Program
Built a scheduling program in shiny R for residents in a dermatology program to automate workflow

Thesis: A Machine Learning Approach to Classifying Australian Communities Vulnerable to Drowning (In process of publication) Jan 2025 - Sep 2025

- Built supervised machine learning models in Python on national drowning data, in collaboration with Royal Life Saving
- Developed predictive classification to identify vulnerable communities and inform prevention strategies

OTHER

Yoga Sculpt Instructor
Corepower/Flowcore

Jan 2022 - Present